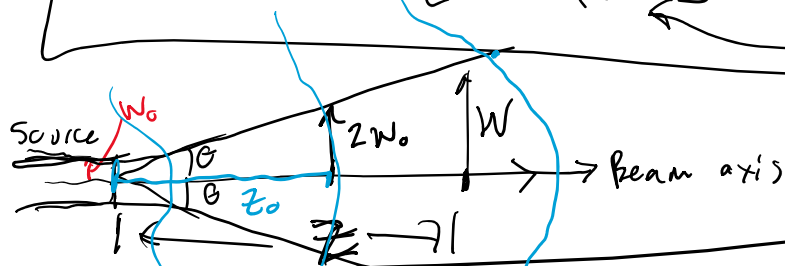


Perfect Gaussian Beams do not exist in real life

$$2w = 2w_0 \left[1 + \left(\frac{z}{z_0} \right)^2 \right]^{1/2} ; z_0 = \frac{\pi(w_0)^2}{\lambda}$$



$$2w = 2w_0 \left[1 + \left(\frac{z\lambda}{\pi(w_0)^2} \right)^2 \right]^{1/2} \rightarrow \text{real beam w/ a Gaussian approximation.}$$

$$2w_0 = w_d$$

$z \gg z_0 \rightarrow$ optical far-field

$\rightarrow$ simplify $2w \approx (2w_0) \left(\frac{z}{z_0} \right) \rightarrow$ linear expansion of the waist.

(beam parameter product) $\boxed{w_0 \theta = \frac{\lambda}{\pi}}$

Non-Perfect Gaussian beams are the Reality

M^2 -factor $\rightarrow$ Perfect Gaussian $M^2 = 1$

initial waist
in real life

anything else

$M^2 > 1$

$$M^2 = \frac{(w_{or})(\theta_R)}{w_0(\theta_0)} = \frac{w_{or}(\theta_R)}{\left(\frac{\lambda}{\pi} \right)}$$

w_{or} - real initial waist (source output radius)

θ_R = Real divergence angle

Realistic approximation

$$2W_R = 2W_0 \left[1 + \left(\frac{z \lambda M^2}{\pi W_0} \right)^2 \right]^{1/2}$$

optical far-field

$$\begin{aligned} 2W_R &\approx M^2(2W_0) \\ \theta_R &= M^2(\theta_0) \end{aligned}$$

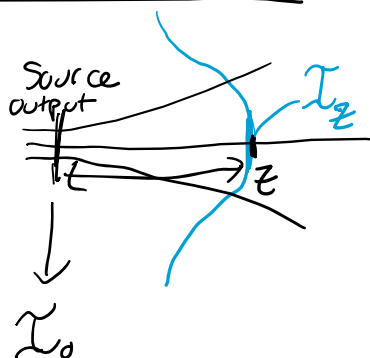
perfect equivalents

$$2W_R \approx M^2 \left(2W_0 \frac{z}{z_0} \right)$$

$$\theta_R = M^2 \left(\frac{\lambda}{W_0 \pi} \right)$$

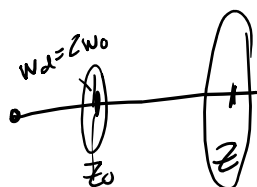
Irradiance ($\mathcal{I}$) of a Realistic Gaussian Beam

$$\mathcal{I}_z(w) = \mathcal{I}_0 \left(\frac{w_0}{w(z)} \right)^2 \left(e^{-\frac{2(w)^2}{(w_0)^2}} \right)$$



@ $z \gg z_0$

$$\mathcal{I}_{axis}(z) = \mathcal{I}_0 \left(\frac{z_0}{z} \right)^2$$



Initial Power at source

$$P_0 = \frac{1}{2} \left[\mathcal{I}_0 (\pi w_0^2) \right] [\text{Watts}]$$

$$P_z = \frac{1}{2} \mathcal{I}_z (\pi (w_z)^2) = P_0 \quad \text{as long as the medium does not absorb or "scatter" the light}$$

$$\mathcal{I}_z = \frac{2 P_0}{(\pi (w_z)^2)}$$

